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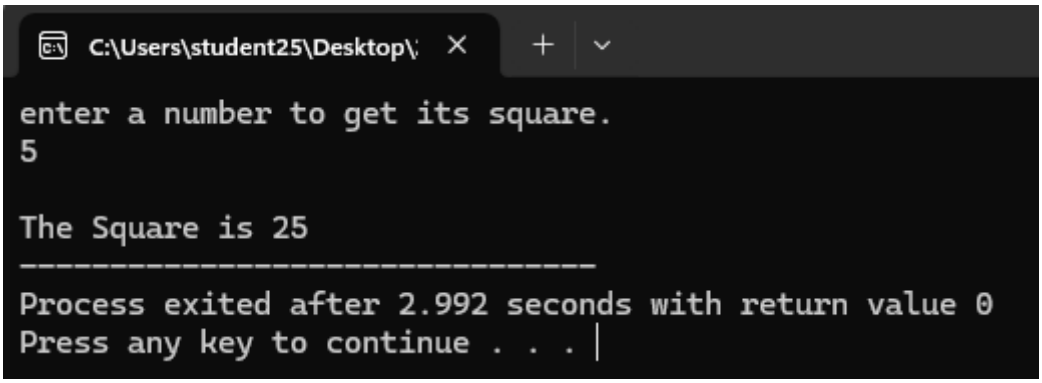
PF Lab 9

4 tasks

Task 1:

```
#include <stdio.h>

int square(int a);

int main()

{

        int a;

        printf("enter a number to get its square.\n");

        scanf("%d", &a);

        printf("\nThe Square is %d", square(a));

return 0;

}

int square(int a)

{

        return a*a;

}
```



Task 2:

```
#include <stdio.h>


int main()
{
	int arr[5];
	for(int j=0;j<5;j++)
	{
		printf("Enter the %d element of array.\n", j+1);
		scanf("%d", &arr[j]);
	}
	int *p = arr;
	for(int i=0;i<5;i++)
	printf("\nAddress of arr[%d] : %d, value: %d\n",i, (p+i), *(p+i));
	return 0;
}
```

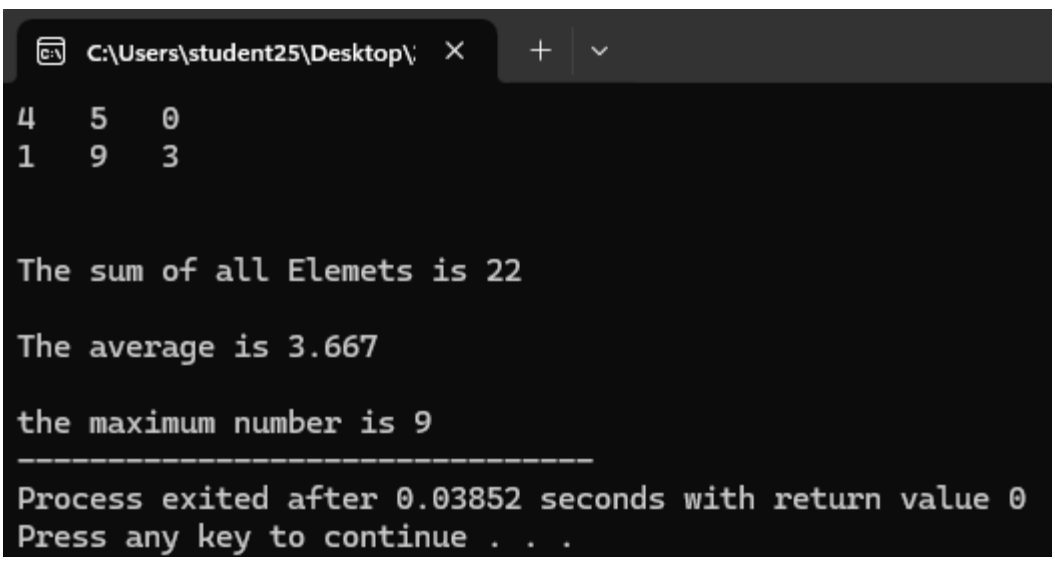
```
C:\Users\student25\Desktop\

Enter the 1 element of array.
10
Enter the 2 element of array.
20
Enter the 3 element of array.
30
Enter the 4 element of array.
40
Enter the 5 element of array.
50

Address of arr[0] : 6487584, value: 10

Address of arr[1] : 6487588, value: 20

Address of arr[2] : 6487592, value: 30

Address of arr[3] : 6487596, value: 40

Address of arr[4] : 6487600, value: 50

--------------------------------
Process exited after 3.345 seconds with return value 0
Press any key to continue . . .
```

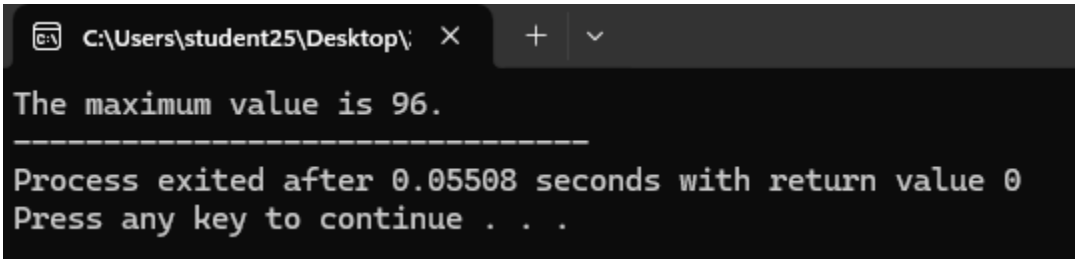
Task 3:

```
#include <stdio.h>
void displayMatrix(int arr[2][3]);
int calculateSum(int arr[2][3]);
float calculateAverage(int arr[2][3]);
int findMax(int arr[2][3]);
float sumg;
int main()
{
	int matrix[2][3] =
	{
		{4,5,0},
		{1,9,3}
	};
	sumg=calculateSum(matrix);
	displayMatrix(matrix);
	printf("\n\n");
	printf("The sum of all Elemets is %d", calculateSum(matrix));
	printf("\n\n");
	printf("The average is %.3f", calculateAverage(matrix));
	printf("\n\n");
	printf("the maximum number is %d", findMax(matrix));
}

void displayMatrix(int arr[2][3])
{
	for(int i=0;i<2;i++)
	{
		for (int j=0; j<3;j++)
```

```
		{
			printf("%d   ",arr[i][j]);
		}
		printf("\n");
	}
}

int calculateSum(int arr[2][3])
{
	int sum=0;
	for(int i=0;i<2;i++)
	{
		for (int j=0; j<3;j++)
		{
			sum+=arr[i][j];
		}
	}
	sumg=sum;
return sum;
}

float calculateAverage(int arr[2][3])
{
	float avg=0;
	avg=(sumg/6);
	return avg;
}

int findMax(int arr[2][3])
```

```
{
        int max=0;
                for(int i=0;i<2;i++)
        {
                for (int j=0; j<3;j++)
                {
                        if(arr[i][j]>max)
                        max=arr[i][j];
                }
        }
return max;
}
```



Task 4:

```
#include <stdio.h>
int findMax(int arr[10]);
int main()
{
	int matrix[10]={12, 85, 73, 94, 63, 25, 96, 19,57, 49};
	printf("The maximum value is %d.", findMax(matrix));
return 0;
}

int findMax(int arr[10])
{
	int max=0;
		for(int i=0;i<10;i++)
	{
		if (arr[i]>max)
		max=arr[i];
	}
return max;
}
```